Sustainable Development Goals & Interlinkages & Nexus Approach

WI 1103 Introduction to the Principles of Sustainability

Week 7



7.1 Sustainable Development Goals & Interlinkages Approach

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Top 10 and Bottom 10 SDG Achievement Rankings

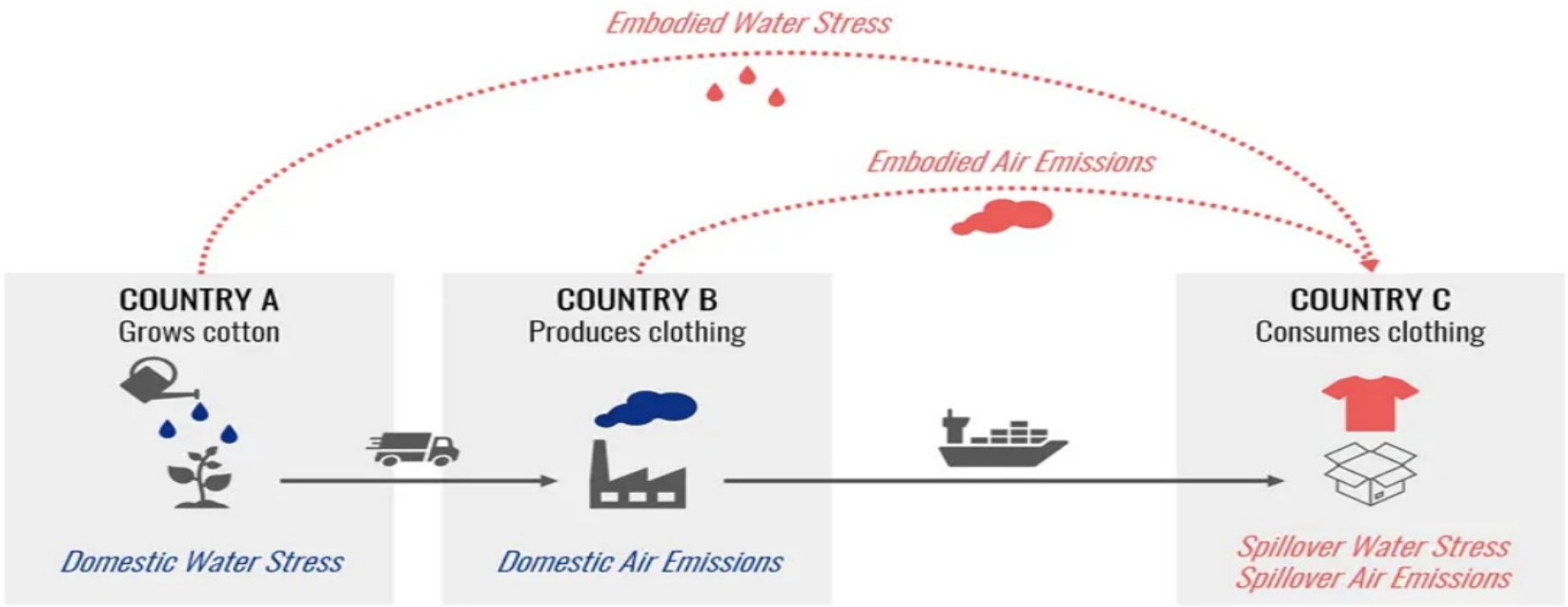
Top 10

Rank	Country	Score	Performance by SDG
1	Finland	87.02	
2	Sweden	85.74	
3	Denmark	85.26	
4	Germany	83.67	
5	France	83.14	
6	Austria	83.01	
7	Norway	82.72	
8	Croatia	82.39	
9	Poland	82.08	
10	Czechia	81.94	

Bottom 10

157	Liberia	52.52	
158	Madagascar	51.05	
159	Niger	50.33	
160	Afghanistan	49.10	
161	Sudan	49.06	
162	Congo, Dem. Rep.	48.24	
163	Yemen, Rep.	47.74	
164	Somalia	46.08	
165	Chad	46.04	
166	Central African Republic	45.21	
167	South Sudan	41.55	

Cross-Border Spillover Effects



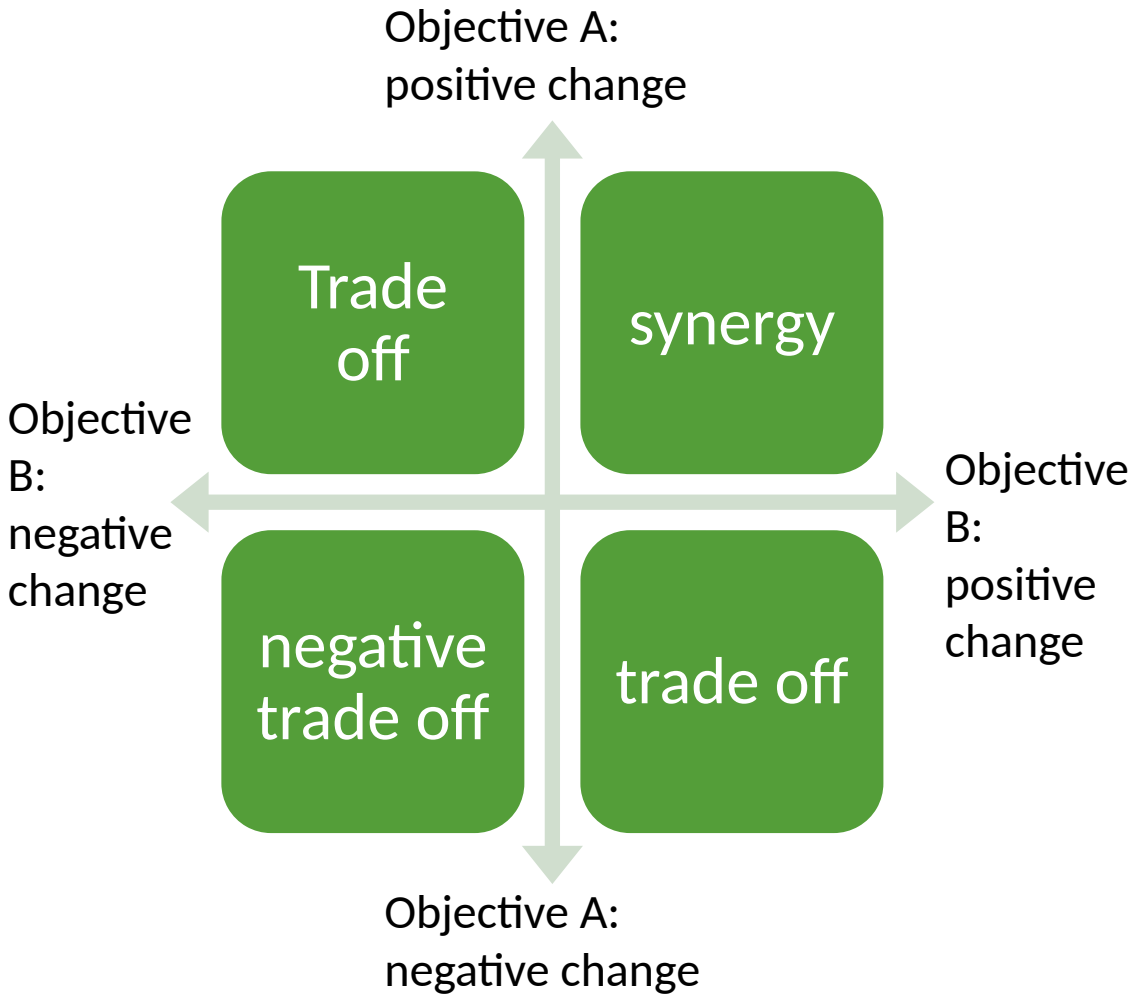
Spillovers effects on the global commons involve impacts generated in various countries at several stages of production and consumption. Learn more about this approach in the Global Commons Stewardship Index.

Negative Spillover Effects of GHGs between Countries

SDG	Type of spillover	Spillover indicator
2 Zero Hunger	Environmental	Export of hazardous pesticides
6 Clean Water & Sanitation	Environmental	Restricted water consumption in imports
8 Decent Work & Economic Growth	Society & Governance	Work accidents in imports
12 Responsible Consumption & Production	Environmental	SO2 emissions in imports
		Nitrogen emissions in imports
		Export of plastic waste
13 Climate Action	Environmental	CO2 emissions in imports
14 Life Below Water	Environmental	Marine Biodiversity (KEHATI) in imports
15 Life on Land	Environmental	Biodiversity (KEHATI) disturbance in imports
16 Peace, Justice and Strong Institutions	Environmental	Export of conventional weapons
17 Partnerships for the Goals	Environmental	International concession
		Corporate tax haven, financial secrecy, profit shifting by Multinational Corps.

Interlinkages between SDGs

- SDGs are not achieved one by one (like an ivory tower) – but rather based on the principle of:
  - Transformative: requiring fundamental changes in how society and the economy are organized
  - Universality: applicable in all countries
  - Indivisibility: integrated within a common framework
- Interlinkages
  - Implementation coherence (context & location specific)
  - Trade-offs: special attention
  - Synergies: accelerating the achievement of SDGs



Activity Characteristics: Relationships between SDGs

Synergies (core activities)

- Creating more positive impact than negative impact on other SDGs

- Types of activities
  Education
  Public health
  Social and legal activities
  Transmission and distribution of renewable energy power plants
  Scientific research

- Strategy: increase positive impact to align with the SDG agenda



Activity Characteristics: Relationships between SDGs

Trade offs (mixed activities)

- Generating both positive and negative impacts on the achievement of SDGs
- Example: Large-scale increase in agricultural production in rural areas (SDG 2)
  - Drivers that increase deforestation (SDG 15), unless there is energy-driven intensification
  - The largest users of freshwater (SDG 6)
  - Reducing the quality of life of local communities ~ inequality (SDG 10)
- Development of renewable energy industries ~ social benefits (energy access) (SDG 7)
  - Long-term trade-offs: bird deaths, or marine habitats, or loss of fertile land (SDG 14)





Activity Characteristics: Relationships between SDGs

Trade offs (opposed activities)

- Activities that have a negative impact on the achievement of SDGs rather than a positive impact
- Include coal and lignite mining, crude oil exploitation, metal mining, and the extraction of building materials (gravel, sand, clay) (SDGs 6, 12, 13, 14, 15).
- Negative impacts on health (SDG 3) arise from the production of alcohol, cigarettes, or soft drinks, which use large volumes of water (SDG 6).





Activity Characteristics: Relationships between SDGs

Trade offs (peripheral activities)

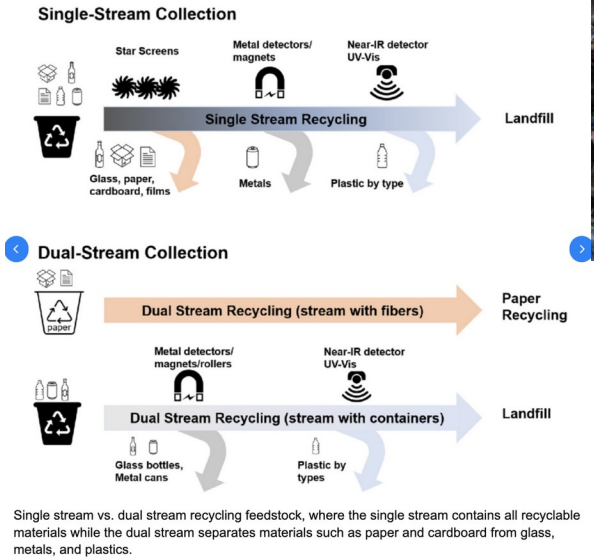
- Activities that do not have a significant positive or negative impact on achieving the SDGs
- Cultural preservation activities through dances, painting, etc.
- Strategy: finding ways to increase the positive impact on achieving the SDGs





In some cases, recycling is more expensive than producing new materials

- Use of recycled materials.
  - The cost of recycling tends to be higher than producing new materials
    - Recycled products are sent to recipient countries, such as China (since 2018, 70% of global plastics are not accepted if below 99% purity level), Indonesia, and African countries
  - Sorting: Single vs dual stream recycling
    - Single stream: collecting recyclable products in one container, mixing various material qualities
    - Dual stream: sorting based on material quality and increasing market value
    - Trade-off: between collection convenience and processing cost/material quality



Single stream vs. dual stream recycling feedstock, where the single stream contains all recyclable materials while the dual stream separates materials such as paper and cardboard from glass, metals, and plastics.

Internasional

China Berhenti Jadi Penampung Sampah, Dunia Kebingungan

Rehia Sebayang, CNBC Indonesia

17 April 2018 14:34





SA beats European plastic recycling rates

Jul 26, 2018 | News

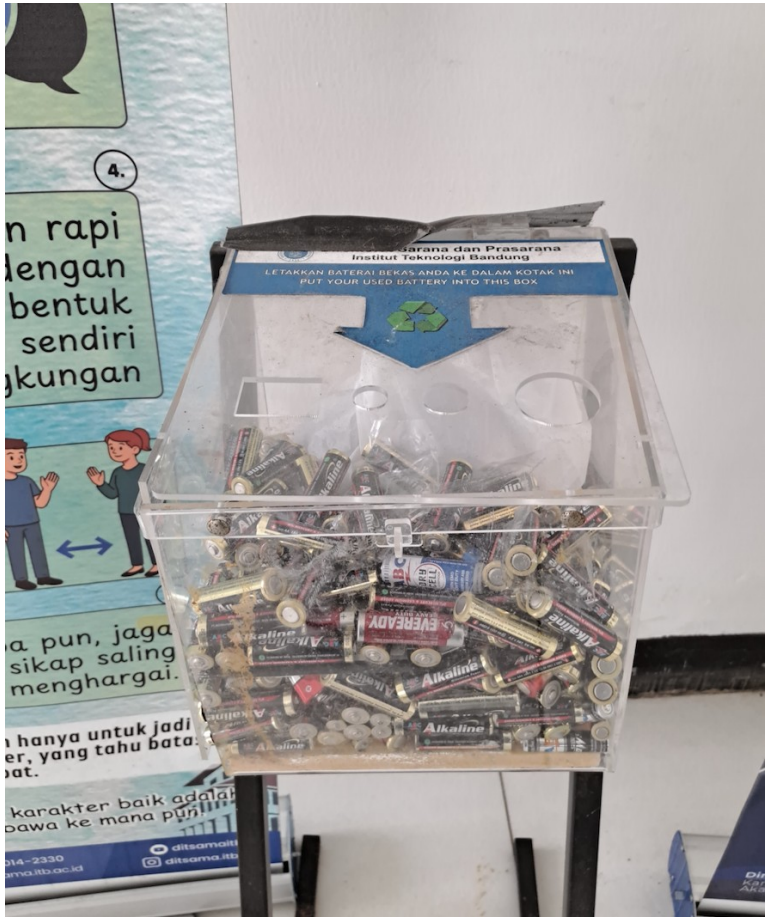
Recycling

- Plastic: Not yet cost-effective (currently)
  - Depends on crude oil prices
- Recycling ~ extends life cycle
- Aluminum: can be recycled with consistent quality
  - Steel: 100% recyclable
  - Glass: recyclable
- Will end up in landfills/incineratorsPaper: can be recycled 6 times ~ then wasted



Beberapa Contoh ‘Recycling’ di Kampus ITB Jatinangor





Sustainable Development Goals & The Role of Technology

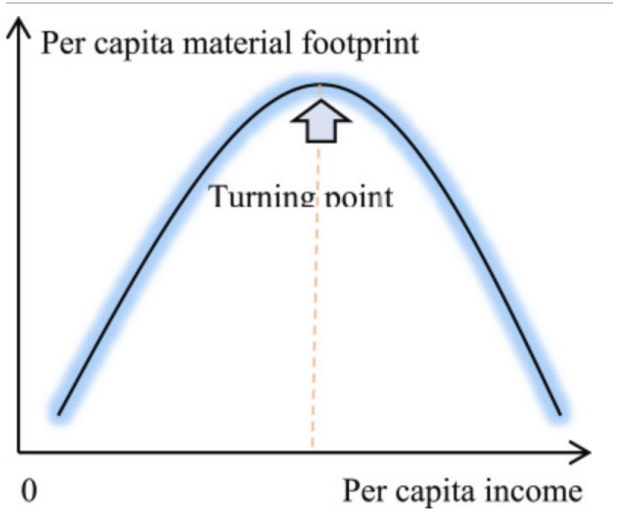
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Relationships among SDGs: The Influence of Technology Use

- Correlated with SDG 12 (Sustainable Production and Consumption)
  - Increasing material/environmental footprints ~ hinders the achievement of SDG 12
  - SDG 12.2, SDG 8.4







Which SDG is often difficult to achieve?

- Often their progress is not reported:
  - SDG 15 Life on land
  - SDG 14 Life below water
  - SDG 5 Gender equality
  - SDG 10 Reduce inequality
- Needs attention:
  - SDG 9 Industry, innovation, and infrastructure
  - SDG 12 Responsible consumption and production
  - SDG 11 Sustainable cities and communities



SDGs Interlinkage approach (Boar, A. et.al.,2021)

- Global north: synergy among SDGs includes all three dimensions (social, economic, environmental)
  - New technology and renewable energy reduce the impact on the environmental dimension
- Global south: synergy among SDGs covers only one dimension

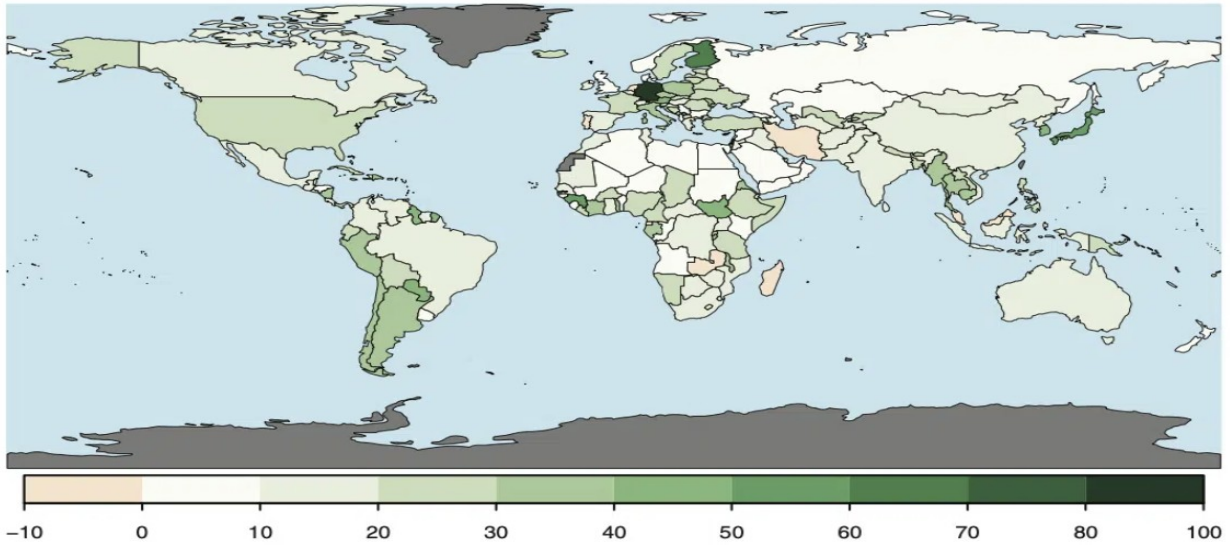
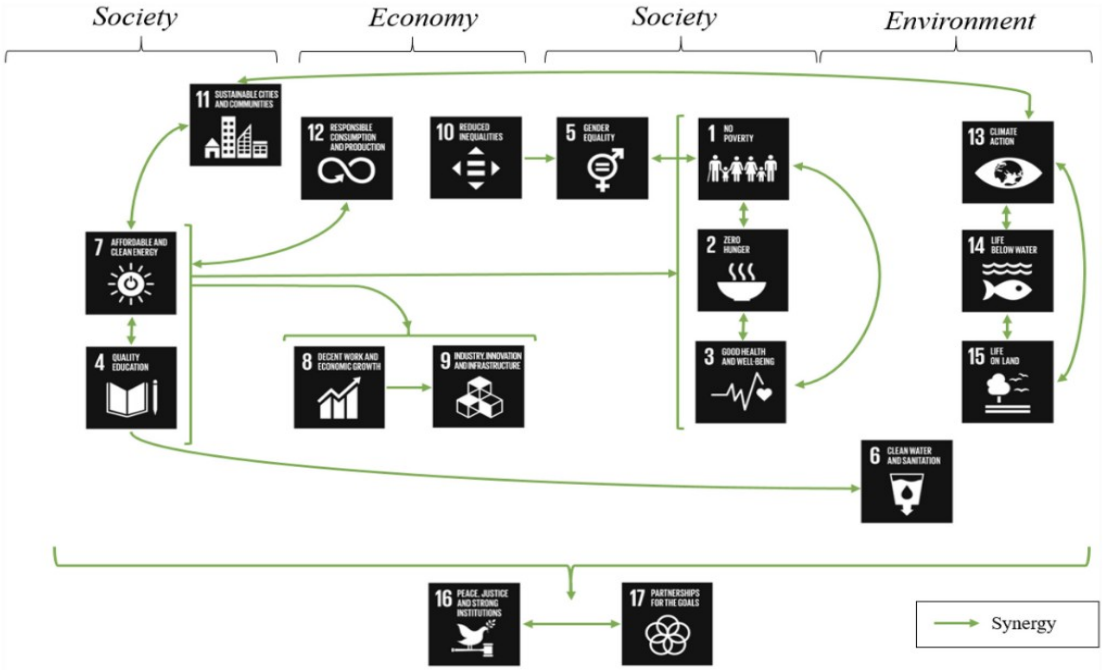


Figure 2: Global distribution of the difference between the shares of synergies and trade-offs among the SDG goals by country. Green colour represents countries where synergies have a higher share than trade-offs, whilst the orange represents the opposite. Source: Pradhan, P., Costa, L., Rybski, D., Lucht, W., & Kropp, J. P, 2017.

Source: https://earth.org/the-interconnectedness-of-sustainable-development-goals/

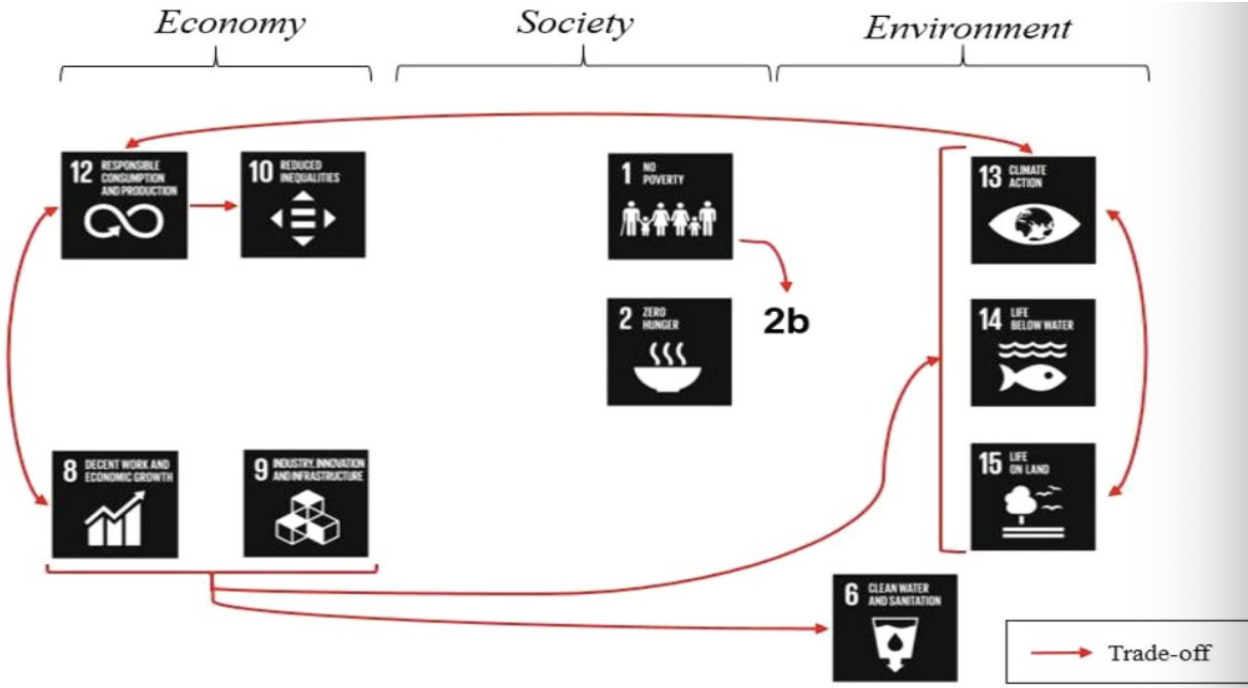
Berdasarkan Tinjauan Pustaka jurnal (Boar. A., et.al., 2021)

Synergy between SDGs



Source(s): Own elaboration

Trade off between SDGs



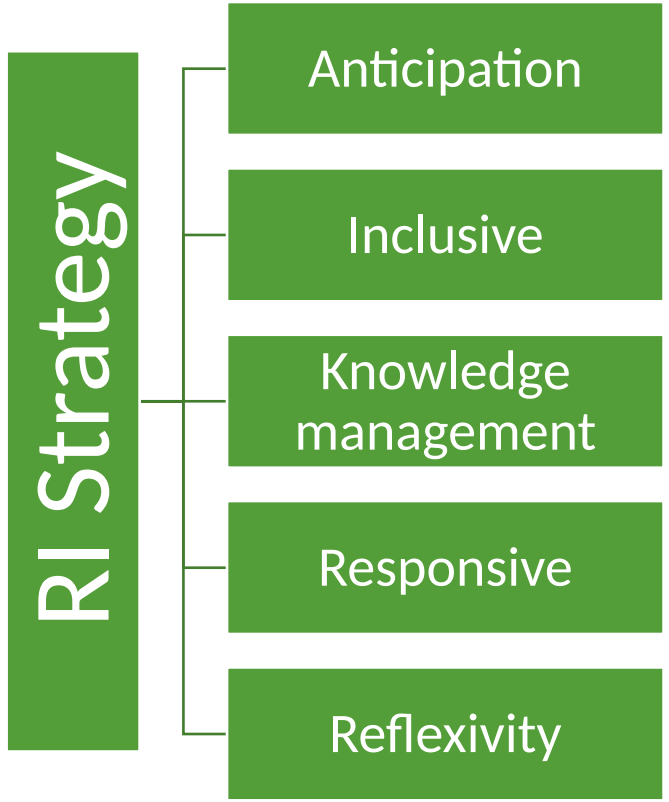
Source(s): Own elaboration

Responsible Innovations (RI)

Responsibility is shown by:

- Avoiding harm
- Doing good

- Long-term trade-offs ~ double effects



Source: Stilgoe et al (2013)

Sustainable Development Goals & Nexus Approach

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Nexus Approach:

- Deepening the integration between goals and targets across sectors, while also clarifying.
  - Avoiding conflicts/overlaps ~ unsustainable & inefficient
  - Strengthening synergy and reducing trade-offs
- Principles:
  - Ensuring access to resources for vulnerable groups
  - Promoting efficient use of resources
  - Ensuring sustainability
- The country prioritizes SDGs, depending on the local context



Food, energy, and water interact with each other and affect the achievement of the SDGs, but they are often addressed separately.



The Food – Energy – Water Nexus approach occurs because:

1

Limited Access

Billions of people worldwide do not have adequate access to food, energy, and water, as part of fulfilling basic needs.

2

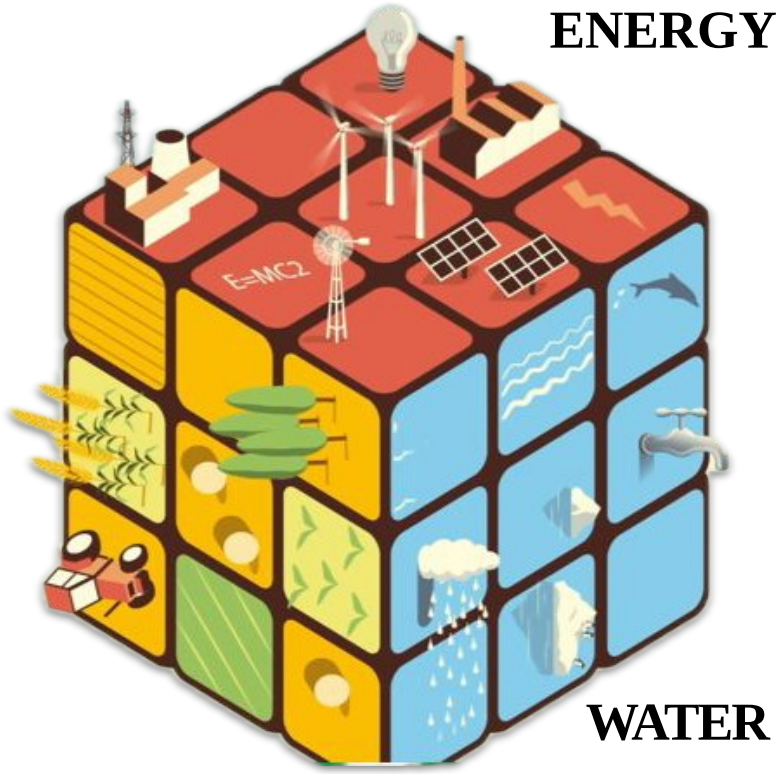
Increasing demand

There has been an increase in global demand in these three areas, due to population growth and the success of development.

3

Limited Resource

Resources for food, water, and energy are limited due to overuse or unsustainable practices.



The Food-Energy-Water Nexus approach was considered for the following reasons:

4

Global Goods

Food, Energy, and Water are among those traded internationally, having a global impact and influencing the economy and policies in the countries that trade them.

5

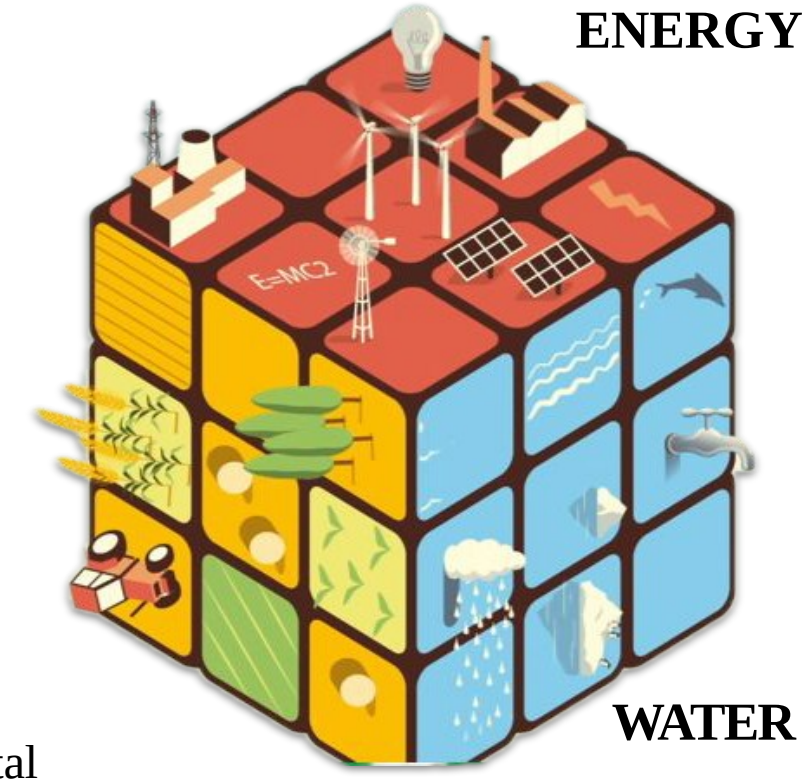
Regional Differences

Regional differences in the provision and variation of supply and demand affect distribution and access

6

Interdependence in Climate and Environmental Change

These three areas are related to climate change and environmental conditions. Climate change affects the availability and quality of all three areas.



The Food-Energy-Water Nexus approach was considered for the following reasons:

7 **Security**

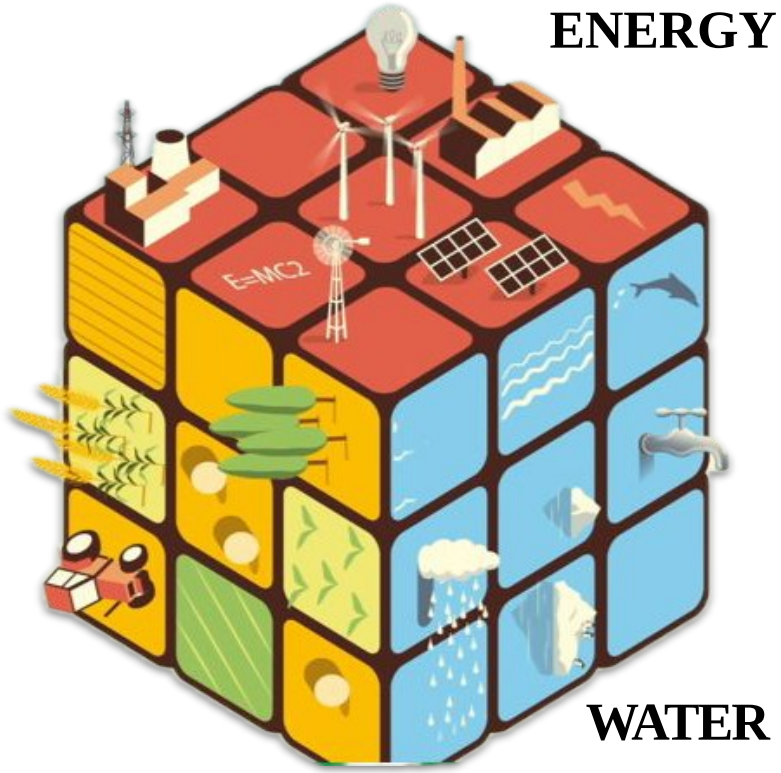
Food, energy, and water are fundamental for society to function. The absence or crisis of any of them results in social instability and economic disruption.

8 **Tightly Regulated Market**

Food, Energy, and water operate in tightly regulated markets, which require caution in policies and regulations.

9 **Identification and Management**

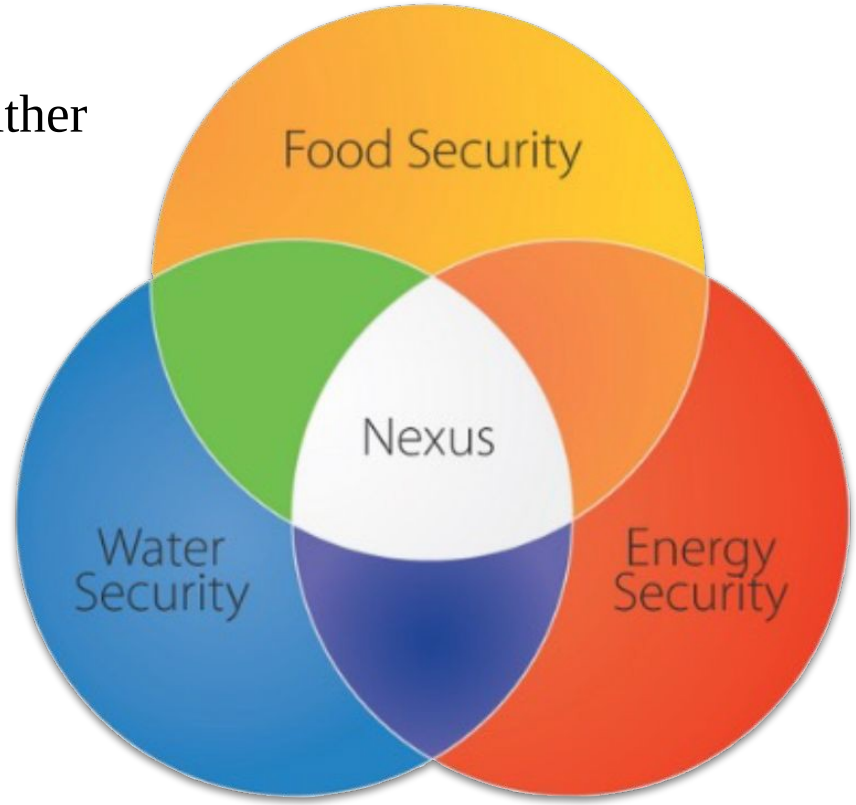
Explicit identification and management of risks will effectively manage these resources, including scarcity, pollution, and unequal access.



Driver Nexus

What are the benefits of the nexus approach?
There are results and impacts, as well as influences that are either direct or indirect, or endogenous and exogenous.

Examples of Driver		
Endogenous	**Exogenous**	
Direct	Direct	Indirect
Resources Governance	Rain	Money Conversion



Examples of Nexus Interventions

Intervention	Water Security	Energy Security	Food Security
Agricultural Conservation	Better soil water retention	Use of engines that can increase fuel consumption	Increase food ingredients and diversity
Energy-efficient cooking stove	Reduced deforestation can increase water retention in the soil	Less energy used for cooking	Change cooking habits and better allocate work time

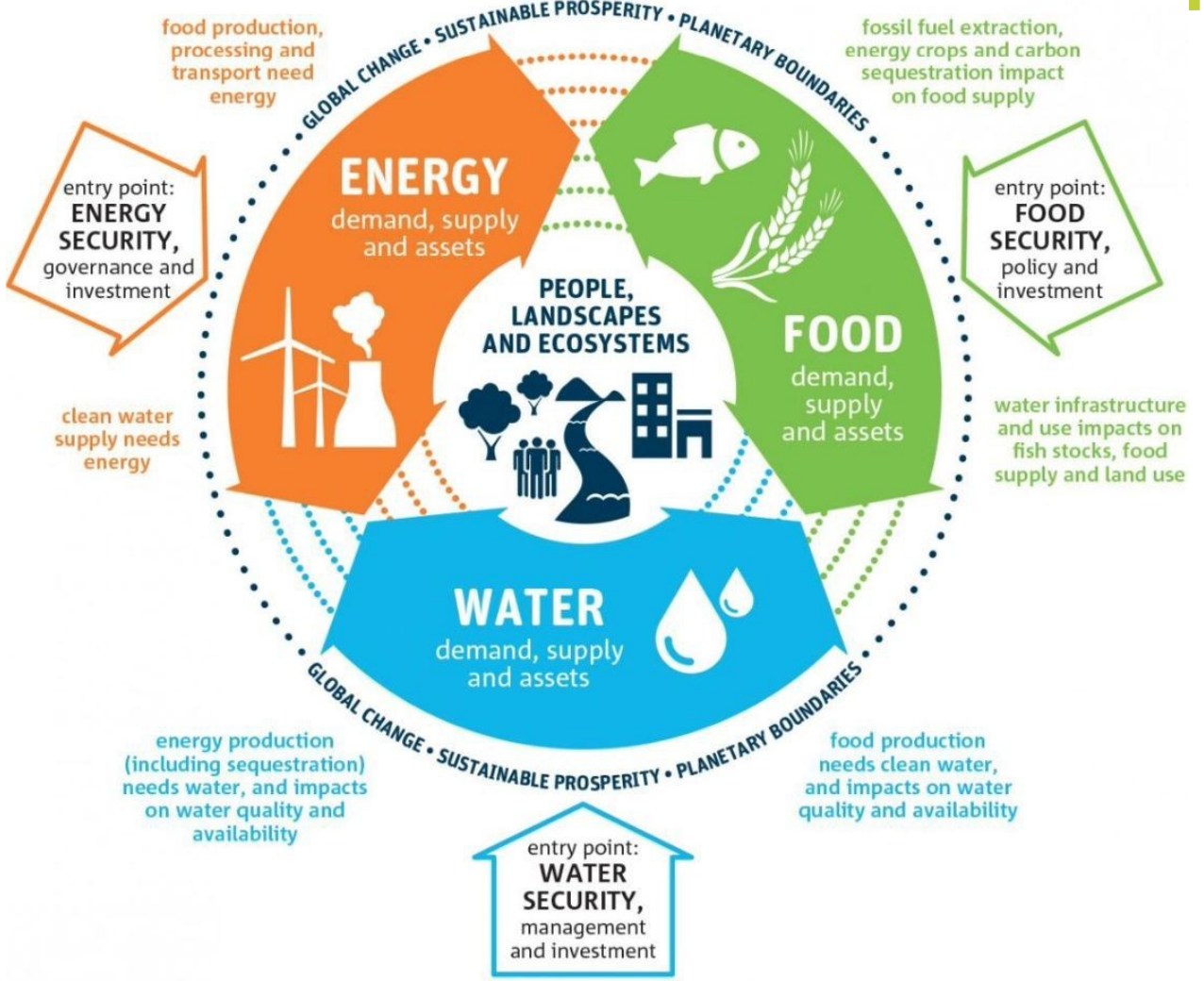
Strategy for Implementing the Nexus Approach

Creating a Nexus Policy

- Develop and approve predetermined goals;
- Share responsibility for achieving predetermined goals;
- Work together to achieve the goal

Set up a meeting to discuss the Nexus

- Continue to coordinate with Ministries not involved;
- Well-organized Ministry responsibilities;
- Horizontal accountability



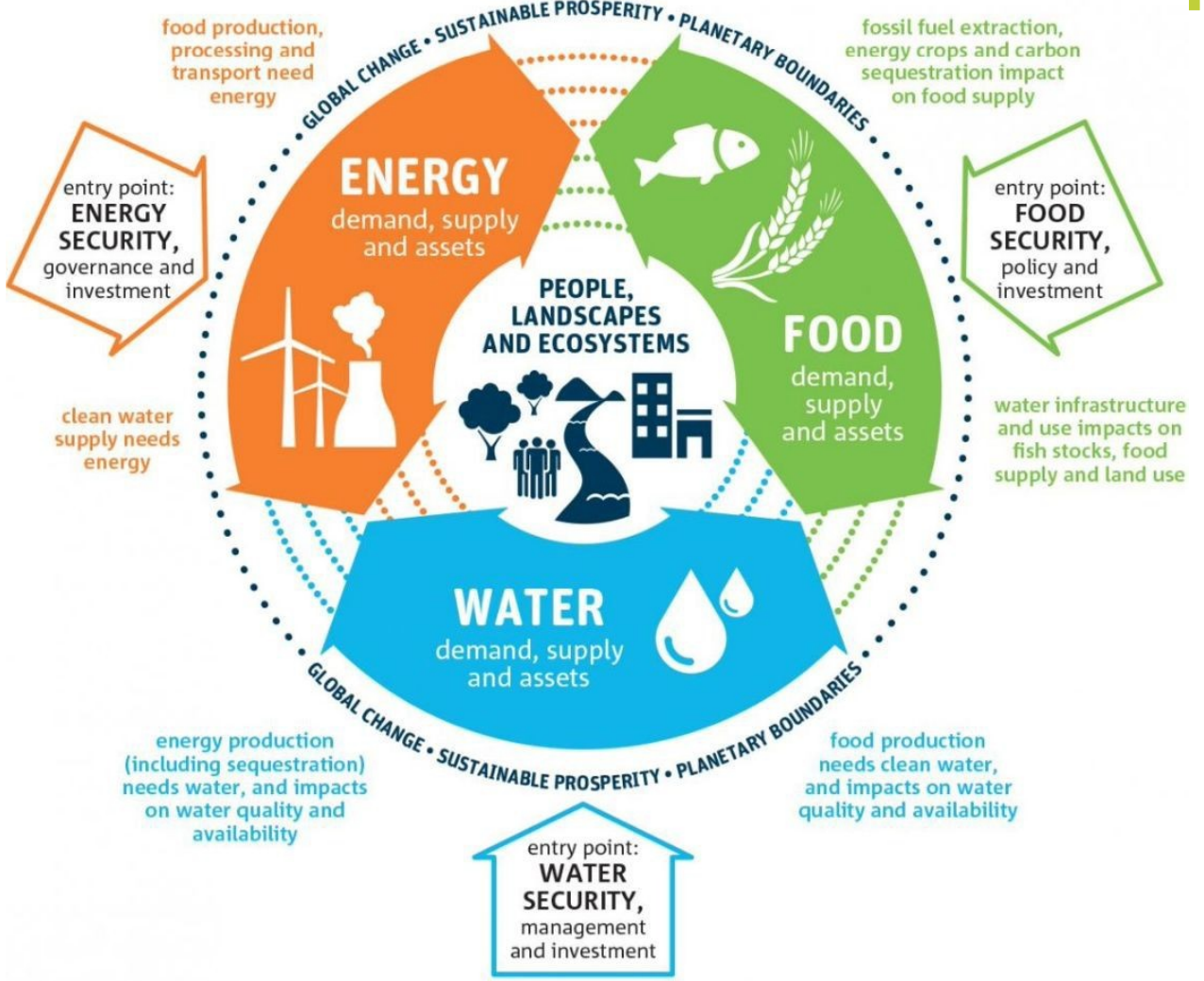
Challenges for Implementing the Nexus Approach

Challenge to researchers

- Lack of framework
- Uncertain time due to group work
- Language barriers
- Problems during publication

Set up a meeting to discuss the Nexus

Incentives when working to solve common problems



Discussions:

- 1. Food, water, energy, and cities
- Imagine that your campus or city wants to expand local food production—such as campus gardens, hydroponics, or urban farming—to strengthen food security. How could this programme affect water and energy use, business costs, access to food for lower-income groups, and the design and social life of the surrounding area?
- 2. Green technology or shifting the problem?
- A city plans to replace fossil-fuel vehicles with electric vehicles, install solar panels, and use digital systems to manage traffic. Does this automatically make the city more sustainable? Who benefits from these policies, and who might bear the hidden environmental or social costs?
- 3. Waste, recycling, and consumer culture
- If a campus aims for "zero waste" by adding separate waste bins and recycling programmes, is that enough to achieve responsible consumption and production? Design a fairer and more effective strategy that addresses waste from the beginning of consumption to its final disposal.

TERIMA KASIH

